

The CMB temperature in the basic form

The eV relation, the ratio to the electron mass and the H_0 forms

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The CMB temperature in the basic form

Abstract

The Casimir-CMB connection is an identity, because the length L_ξ is fixed solely through the CMB temperature (R136). This document collects all routes meant to obtain T_{CMB} from ξ without this circularity and recalculates them in the basic form: natural units with $\alpha = 1$, a single anchor m_e . The eV relation $T_{\text{CMB}} = \frac{16}{9}\xi$ eV with second-order correction agrees to 2×10^{-6} , but holds only in the unit eV, whose size was fixed in 1881 and set exactly in 2019. In the basic form, T_{CMB}/m_e depends on a single number. The candidate $T_{\text{CMB}}/m_e = (8\pi)^{1/4} \xi^{5/2}$ meets the FIRAS value to 0.12σ , together with the corpus form of H_0 gives $T_{\text{CMB}}^4 = 16 H_0 m_e^3$, and makes L_ξ independent of the CMB. The H_0 form with exponent 10 fits it; the older one with $41/4$ is off by 11σ . Neither the prefactor nor the exponent is derived; what remains open is listed at the end.

Status markers

- [B]** Proven algebraically.
- [K]** Confirmed numerically (verification script, comparison with measured values).
- [S]** Setting, open or speculative.
- [X]** Refuted or not viable.
- [Q]** Convention.

Measured values: $T_{\text{CMB}} = 2.72548 \pm 0.00057$ K (FIRAS, Fixsen 2009; relative uncertainty 0.021 %), CODATA 2022. Verification script: `pruef_388_cmb_grundform.py`.

1 Starting point

The length $L_\xi = (15\xi/\pi^2)^{1/4} \hbar c/(k_B T_{\text{CMB}}) = 100.24 \mu\text{m}$ is chosen such that the photon density of the CMB equals ξ/L_ξ^4 there. The comparison with the Casimir energy density at the same length is therefore an identity, not a measurement (Doc. 279, R136). An H_0 anchor does not change this either, since L_ξ depends only on T_{CMB} .

The ΛCDM model does not derive T_{CMB} either. It takes the FIRAS value as a fixed input, and the Planck value $H_0 = 67.4$ km/s/Mpc is a derived quantity that presupposes it. The question is therefore not whether FFGFT is worse off than ΛCDM , but whether it can do more: T_{CMB} as a ratio from ξ , with the same anchor as the masses.

2 The eV relation

Since 2025 the corpus has carried the relation $T_{\text{CMB}} = \frac{16}{9}\xi$ eV, later with a second-order correction (P11):

$$k_B T_{\text{CMB}} = \frac{16}{9}\xi \left(1 - \frac{275}{4}\xi\right) \text{ eV}. \quad (1)$$

In SI, the first order gives 3.7978×10^{-23} J, i.e. 2.7507 K and +0.93 %; the second order gives 2.72549 K and +0.0002 % **[K]**. The agreement is real, its meaning is not: three observations show that it depends on the unit.

First, the relation holds only in eV. Read $\frac{16}{9}\xi$ in meV, in kelvin or in units of m_e , and it is off by factors of 10^3 , 10^4 or 5×10^5 **[K]**. Written as a ratio it reads $T_{\text{CMB}}/m_e = \frac{16}{9}\xi/510\,999$; the number 510 999 is m_e in eV, i.e. the anchor itself, not an expression in ξ .

Second, redefining α changes nothing. With $\alpha = 1$ the charge unit grows by $1/\sqrt{\alpha} = 11.706$ and the potential shrinks by the same factor; energies stay the same **[B]**. T_{CMB} contains no charge, so α can neither create nor remove the relation.

Third, the eV is set, not measured **[Q]**. Its size goes back to the practical volt of 1881 (10^8 electromagnetic CGS units, chosen after the voltage of the Daniell cell), and thus to the centimetre, gram and second. Every later stage – the international volt of 1908, absolute units in 1948, the Josephson constant in 1990 – kept this size. Since 20 May 2019, with the fixed elementary charge, exactly $1 \text{ eV} = 1.602176634 \times 10^{-19}$ J (SI Brochure, 9th ed.). The CMB was discovered in 1965; it cannot have entered the size of the eV.

Then there is the coefficient. Exactly required would be 68.766; $275/4 = 68.75$ is the nearest fraction with denominator 4. Fractions $n/4$ lie 3.3×10^{-5} apart in T , so a fitting one can always be found to within $\pm 1.7 \times 10^{-5}$. The second order lies 0.01σ from the central value. Beyond the measurement precision an agreement proves nothing; a correct formula would typically lie about 1σ off, and it comes this close only with a probability of about 1 %. This suggests that

the coefficient was chosen at the central value. The classification of P11 in R137 (i) stands: unit-dependent agreement, coefficient not derived **[S]**.

For the bookkeeping this means: a formula of the form "quantity = number times eV" uses the unit eV as an anchor. That is allowed as long as it is the only one. The mass ladder, however, uses m_e ; both together require that m_e/eV follow from ξ . The only proximity is $\text{eV}/m_e = 1.271 \xi^{3/2}$, 0.17 % from $(4/\pi) \xi^{3/2}$ **[K]**; that would derive the size of the volt from ξ and is not viable **[X]**.

3 Other reference energies

The basic form fixes an energy unit of its own: from $\xi E_0^2 = 1$ follows $u = \sqrt{\xi} E_0$, so with $E_0 = \sqrt{m_e m_\mu}$, $u = 84.85 \text{ keV}$. In this unit $T_{\text{CMB}} = 2.768 \times 10^{-9} u = 0.0876 \cdot \frac{16}{9} \xi^2 u$ **[K]**, without a clean prefactor. Referred to the energy scale of the atom, $T_{\text{CMB}}/(\alpha^2 m_e) = 0.0647 \xi$ **[K]**, likewise without a clean prefactor. Neither reference yields a relation.

4 The basic form: T_{CMB}/m_e

In natural units T_{CMB} depends on the Hubble rate, the Planck energy and the share of photons in the critical density. With $\rho_Y = \frac{\pi^2}{15} T^4$ and $\rho_{\text{crit}} = 3H_0^2 E_P^2/(8\pi)$,

$$\frac{T_{\text{CMB}}}{m_e} = \left(\frac{45 \Omega_Y}{8\pi^3} \right)^{1/4} \left(\frac{H_0 E_P}{m_e^2} \right)^{1/2}. \quad (2)$$

With the corpus forms $H_0 = \frac{\pi}{2} \xi^{10} m_e$ (Docs. 165, 279) and $m_e/E_P = \frac{4/3}{5\pi} \xi^{11/2}$ (Doc. 383), $H_0 E_P/m_e^2 = \frac{15\pi^2}{8} \xi^{9/2}$ **[B]**. If Ω_Y is proportional to ξ , then $T_{\text{CMB}}/m_e \propto \xi^{5/2}$. Measured:

$$\frac{T_{\text{CMB}}}{m_e} = K \xi^{5/2}, \quad K = 2.23897 \quad \textbf{[K]}. \quad (3)$$

This explains the exponent 2.41 read off without a prefactor. The question is thereby reduced to one number, equivalent to Ω_Y/ξ .

Simple candidates lie outside the measurement precision: $\sqrt{5}$ (−0.13 %), $9/4$ (+0.49 %), $64/(9\pi)$ (+1.10 %) **[X]**. Inside lies

$$\boxed{\frac{T_{\text{CMB}}}{m_e} = (8\pi)^{1/4} \xi^{5/2}} \quad K = 2.23903, +0.0025 \%, 0.12\sigma \quad \textbf{[K]}. \quad (4)$$

8π is the coupling in $8\pi G$, which motivates the candidate but does not justify it **[S]**.

As a check against chance, all 3370 expressions $(p/q \cdot \pi^k)^{1/n}$ with $p, q \leq 30$, $|k| \leq 3$, $n \leq 4$ in the range 1.5 to 3.5 were searched. Three fall within FIRAS; 1.6 are expected by chance **[K]**. Restricted to $p, q \leq 10$ (383 expressions), only $(8\pi)^{1/4}$ remains, with 0.2 hits expected by chance. The candidate is thus the simplest, but not the only one; a coincidence is not excluded.

5 Consequences of the candidate

Relation to H_0 . Equivalently $T_{\text{CMB}}^4 = 8\pi \xi^{10} m_e^4$, with the same exponent 10 as in H_0 . Together with $H_0 = \frac{\pi}{2} \xi^{10} m_e$, ξ drops out:

$$\boxed{T_{\text{CMB}}^4 = 16 H_0 m_e^3} \quad \textbf{[B]}. \quad (5)$$

From the measured T_{CMB} and m_e follows $H_0 = 66.81 \pm 0.06$ km/s/Mpc **[K]**. This lies 1.2σ from Planck (67.4 ± 0.5) and far from SH0ES. The uncertainty comes from FIRAS alone and is about eight times smaller than that of direct H_0 measurements.

Photon density. With the T0 values of H_0 and E_P , $\Omega_\gamma = \frac{4096}{10125} \xi = 0.4045 \xi$ **[B]**; with the measured E_P instead of the T0 form, 0.4154ξ .

L_ξ without the CMB. Inserting the candidate into the definition of L_ξ , T_{CMB} drops out:

$$L_\xi = \left(\frac{15}{8\pi^3} \right)^{1/4} \xi^{-9/4} \bar{\lambda}_e = 100.24 \mu\text{m} \quad \textbf{[K]}. \quad (6)$$

L_ξ is thus a length built from ξ and the Compton wavelength of the electron. The ratio $\pi^2/(720\xi) = 102.8$ between the Casimir and CMB densities at L_ξ remains an identity; the circularity, however, moves from L_ξ to the prefactor $(8\pi)^{1/4}$.

6 The H_0 forms compared

The corpus contains two forms: $E_H = E_0 \xi^{41/4}$ (Docs. 026, 064; exponent calibrated to the measured value, P35) and $H_0 = \frac{\pi}{2} \xi^{10} m_e$ (Docs. 165, 279). They differ by 1%, since $E_0 \xi^{1/4} = 0.9904 \cdot \frac{\pi}{2} m_e$ **[K]**. In the H_0 measurement this goes unnoticed; T_{CMB} , however, is measured more than thirty times more precisely and separates the forms via $T^4 = 16 H_0 m_e^3$:

H_0 form	H_0 (km/s/Mpc)	T_{CMB} (K)	Deviation
$\frac{\pi}{2} \xi^{10} m_e$	66.821	2.72555	+0.003 %, 0.1σ [K]
$E_0 \xi^{41/4}$, $E_0 = 7.398$ MeV	66.179	2.71898	−0.24 %, 11σ [X]
$E_0 \xi^{41/4}$, $E_0 = \sqrt{m_e m_\mu}$	65.731	2.71436	−0.41 %, 20σ [X]
backwards from T_{CMB}	66.81 ± 0.06	measured	–

If the candidate holds, the form with $41/4$ is excluded by the CMB, and the transition to the exponent 10 in Doc. 165 is confirmed. This confirmation is not independent: the prefactor $(8\pi)^{1/4}$ was found at the measured value, and the formula already contains ξ^{10} . What is shown is that the same ξ^{10} structure carries both H_0 and T_{CMB} , with the simple prefactors $\pi/2$ and 8π .

7 Comparison

Route	Anchor	Agreement	Classification
$\frac{16}{9} \xi$ eV	eV	+0.93 %	unit-dependent [S]
$\frac{16}{9} \xi (1 - \frac{275}{4} \xi)$ eV	eV	2×10^{-6}	unit-dependent, fitted [S]
T/u , $u = \sqrt{\xi} E_0$	m_e	no prefactor	no relation
$T/(\alpha^2 m_e)$	m_e	no prefactor	no relation
$\sqrt{5}$, $9/4$, $64/(9\pi)$	m_e	0.13 to 1.1 %	outside FIRAS [X]
$(8\pi)^{1/4} \xi^{5/2}$	m_e	0.12σ	candidate [S]
ΛCDM	T_{CMB} itself	–	measured value as input [Q]

8 What remains open

The circularity of L_ξ is resolved only conditionally: it has been moved to one number. Open are

- the prefactor $(8\pi)^{1/4}$, equivalent to $\Omega_\gamma = \frac{4096}{10125}\xi$; motivated by $8\pi G$, not derived; with 1.6 hits expected in the wide search space, a coincidence is not excluded **[S]**;
- the exponent 10 and the prefactor $\pi/2$ of H_0 (P20); the T relation inherits them, since $5/2 = 10/4$ **[S]**;
- why the photon density today is a fixed fraction of ξ ; in an expanding model this would be a coincidence of epoch, in a static one a statement that needs a derivation **[S]**;
- the eV relation P11: unit-dependent, coefficient $275/4$ fitted; it remains as a historical agreement but contributes nothing to the basic form (R137, R70) **[S]**.

The candidate can be tested through H_0 : it requires $H_0 = 66.81 \pm 0.06$ km/s/Mpc. A model-independent H_0 measurement with a precision of a few tenths of a per cent would confirm or exclude it.

9 Conclusion

The eV relation agrees because the eV brings the electron mass along as a second anchor; in every unit that FFGFT fixes itself, the agreement disappears. In the basic form, T_{CMB}/m_e depends on a single number, and the simplest candidate, $(8\pi)^{1/4}\xi^{5/2}$, meets FIRAS to 0.12σ . It links T_{CMB} and H_0 as $T^4 = 16 H_0 m_e^3$, fixes L_ξ without the CMB, and excludes the old H_0 form with $41/4$. It is not derived; the prefactor and, as for H_0 , the exponent 10 remain open.

Verification script: 2/python/Dok388_Skripte/pruef_388_cmb_grundform.py — 27/27.